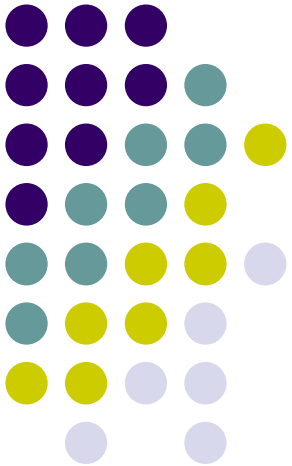
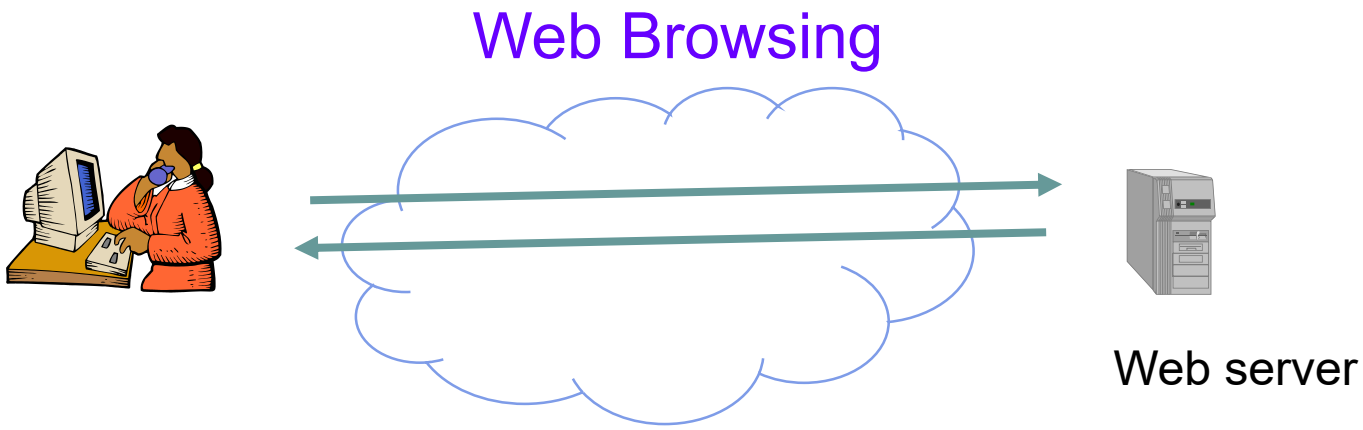
Introduction

Contain slides by Leon-Garcia and Widjaja

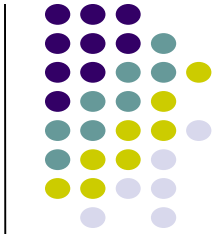


Communication Services & Applications

- Communication network: a set of equipment and facilities that provide a service
- A communication service enables the exchange of information between users at different locations.
- Communication services & applications are everywhere.

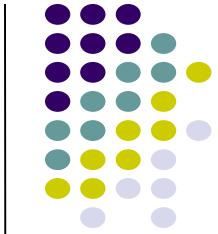


Many other examples!



- Peer-to-peer applications
  - Napster, Gnutella, Kazaa file exchange
  - Searching for ExtraTerrestrial Intelligence (SETI)
- Audio & video streaming
- Network games
- On-line purchasing
- Text messaging in PDAs, cell phones (SMS)
- Voice-over-Internet

Services & Applications



- Service: Basic information transfer capability
  - Internet transfer of individual block of information
  - Internet reliable transfer of a stream of bytes
  - Real-time transfer of a voice signal
- Applications build on communication services
  - E-mail & web build on reliable stream service
  - Fax and modems build on basic telephone service
- New applications build on multiple networks
  - SMS builds on Internet reliable stream service and cellular telephone text messaging

What is a communication network?



- The equipment (hardware & software) and facilities that provide the basic communication service
- Virtually invisible to the user; Usually represented by a cloud

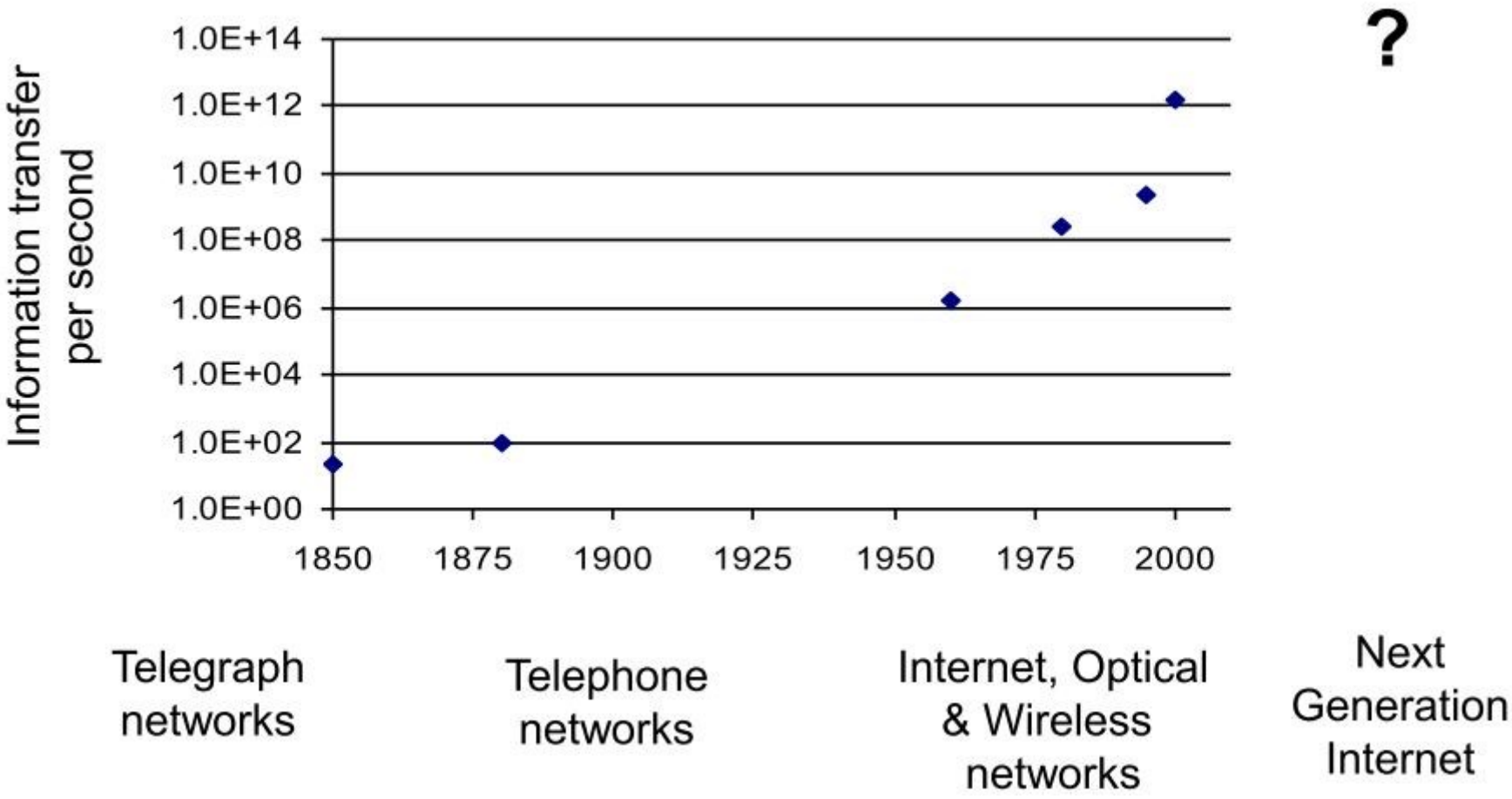
- **Equipment**
  - Routers, servers, switches, multiplexers, hubs, modems, …
- **Facilities**
  - Copper wires, coaxial cables, optical fiber
  - Ducts, conduits, telephone poles …

Communication Network Architecture



- Network Architecture: the plan that specifies how the network is built and operated
- Architecture is driven by the network services
- Overall communication process is complex
- Network architecture partitions overall communication process into separate functional areas called layers
- Three network architectures:
  - telegraph,
  - telephone, and
  - computer networks

Network Architecture Evolution



Network Architecture Evolution

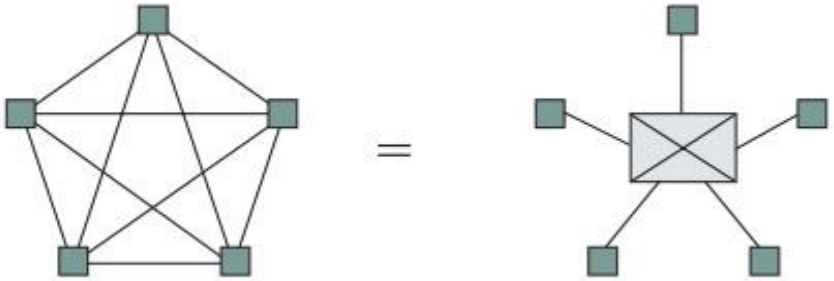


- Telegraph Networks
  - Message switching & digital transmission
- Telephone Networks
  - Circuit Switching
  - Analog transmission → digital transmission
  - Mobile communications
- Internet
  - Packet switching & computer applications
- Next-Generation Internet
  - Multiservice packet switching network

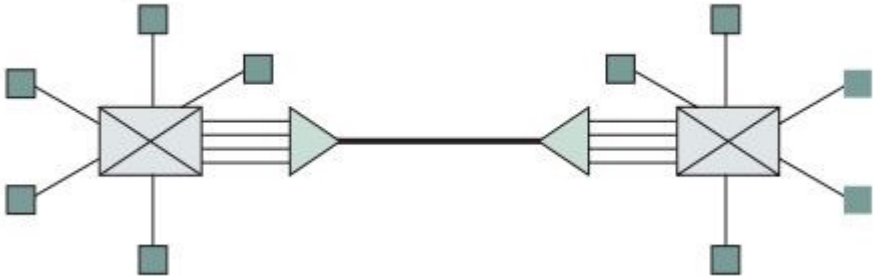
Switch and Multiplexer

- Switch:
  - Routing: decide the exit point
  - Forwarding: move the information
- Multiplexer:
  - ‘trunk’ to interconnect switches

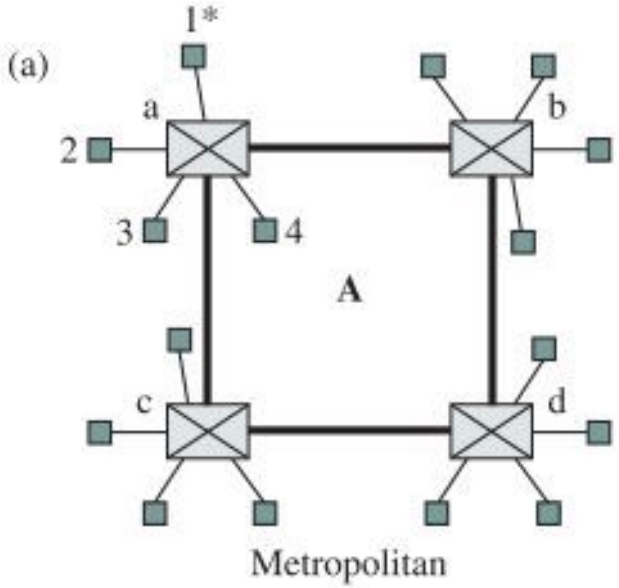
(a) A switch provides the network to a cluster of users.



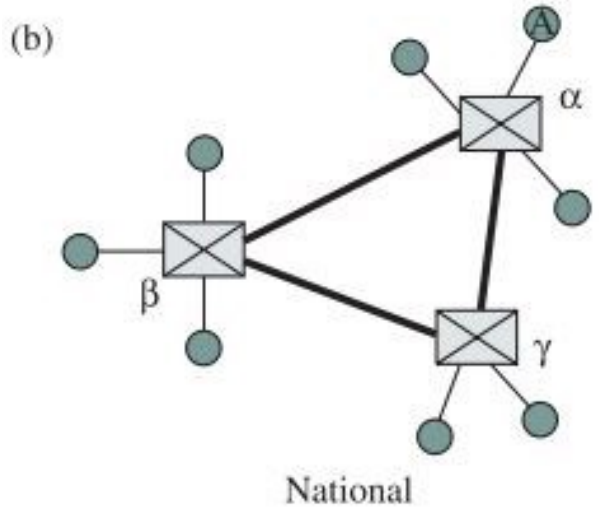
(b) A multiplexer connects two access networks.



Hierarchical Network Topology



Metropolitan network **A** consists of access subnetworks a, b, c, d.



National network consists of regional subnetworks α, β, γ.

Metropolitan network **A** is part of regional subnetwork α.

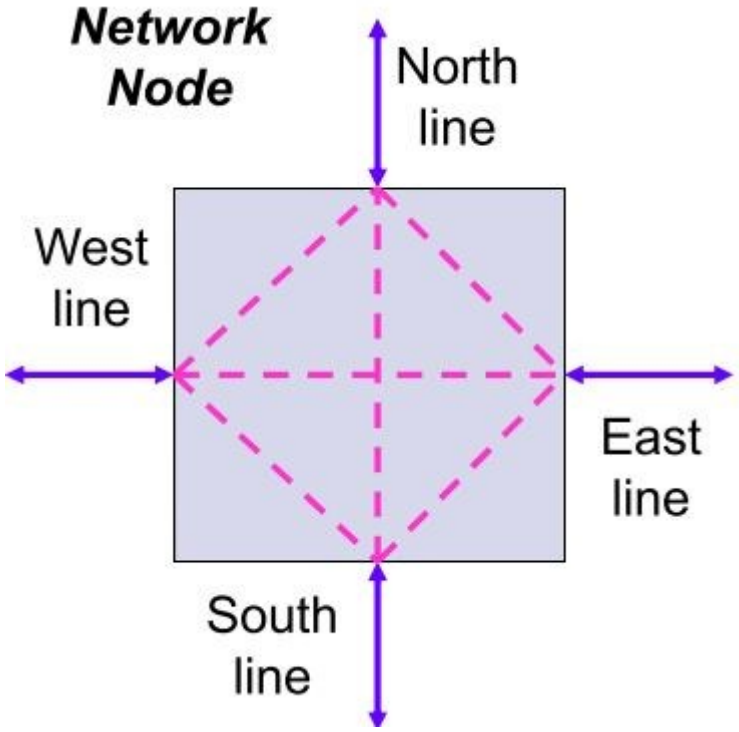
The Essential Network Functions



- Basic user service—the primary service or services that the network provides to its users.
- Switching approach—the means of transferring information flows between communication lines.
- Terminal—the end system that connects to the network.
- Information representation—the format of the information handled by the network.
- Transmission system—the means for transmitting information across a physical medium.
- Addressing—the means for identifying points of connection to the network.
- Routing—the means for determining the path across the network.
- Multiplexing—the means for connecting multiple information flows into shared connection lines.

Message Switching in Telegraph

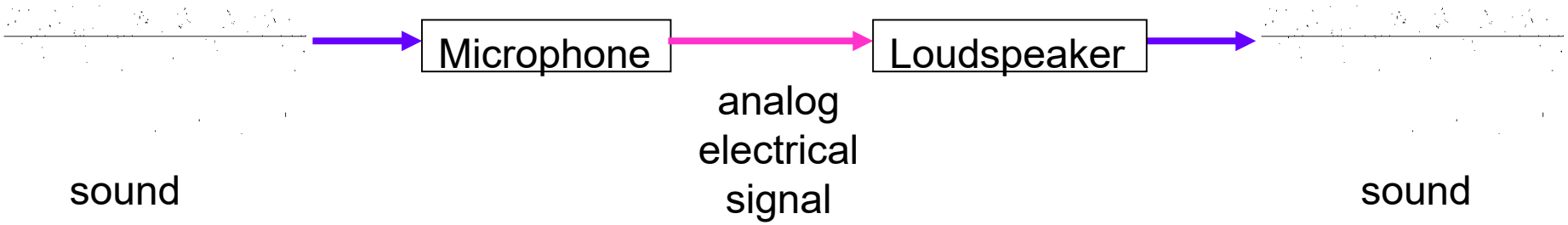
- Network nodes were created where several optical telegraph lines met (Paris and other sites)
- Store-and-Forward Operation:
  - Messages arriving on each line were decoded
  - Next-hop in route determined by destination address of a message
  - Each message was carried by hand to next line, and stored until operator became available for next transmission



Bell's Telephone

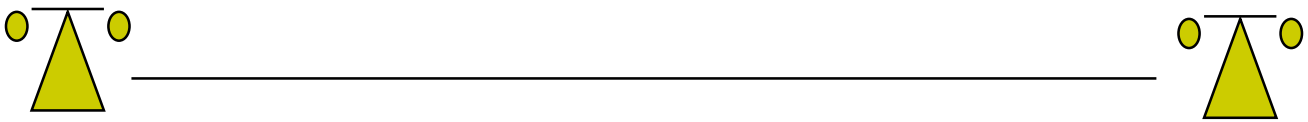
- Alexander Graham Bell (1875) working on harmonic telegraph to multiplex telegraph signals
- Discovered voice signals can be transmitted directly
  - Microphone converts voice pressure variation (sound) into *analogous* electrical signal
  - Loudspeaker converts electrical signal back into sound
- Telephone patent granted in 1876
- Bell Telephone Company founded in 1877

Signal for "ae" as in cat



Signaling

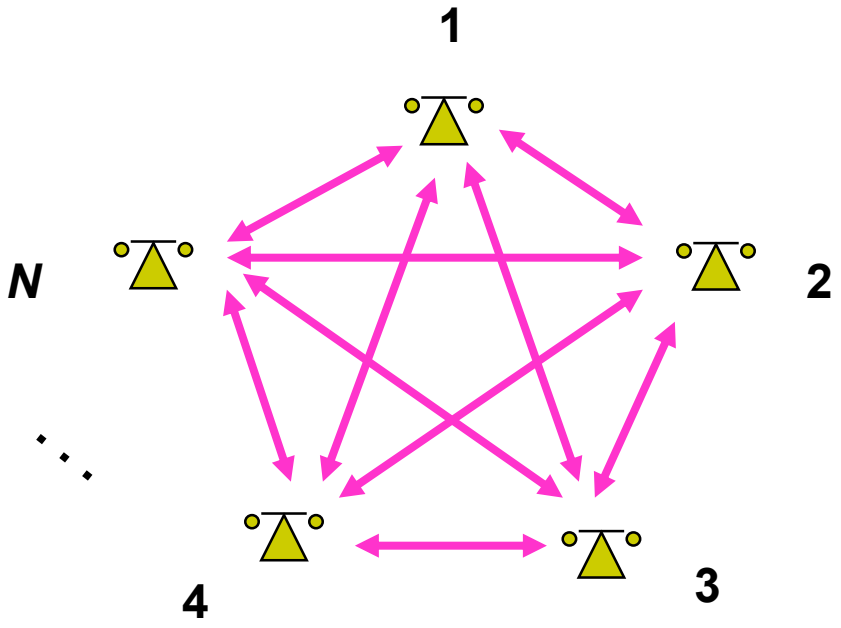
- Signaling required to establish a call
  - Flashing light and ringing devices to alert the called party of incoming call
  - Called party information to operator to establish calls



Signaling + voice signal transfer

The N^2 Problem

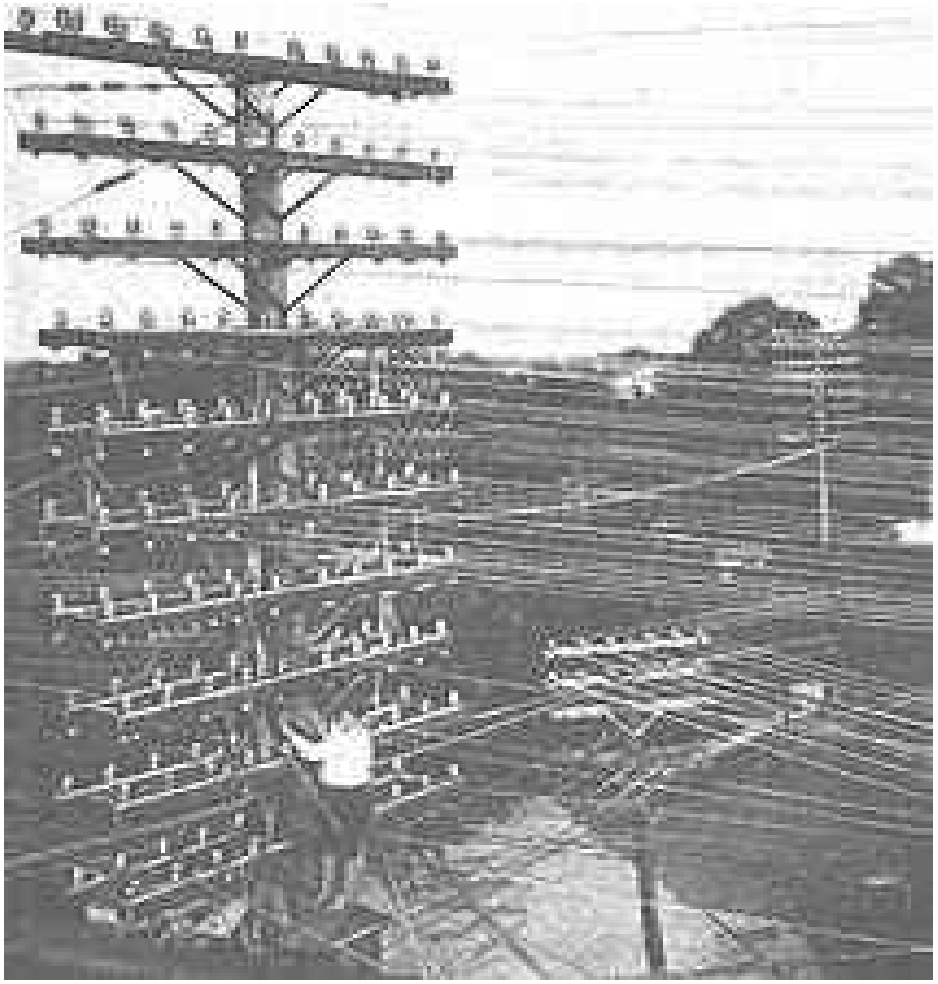
- For N users to be fully connected *directly*
- Requires $N(N-1)/2$ connections
- Requires too much space for cables
- Inefficient & costly since connections not always on



$N = 1000$

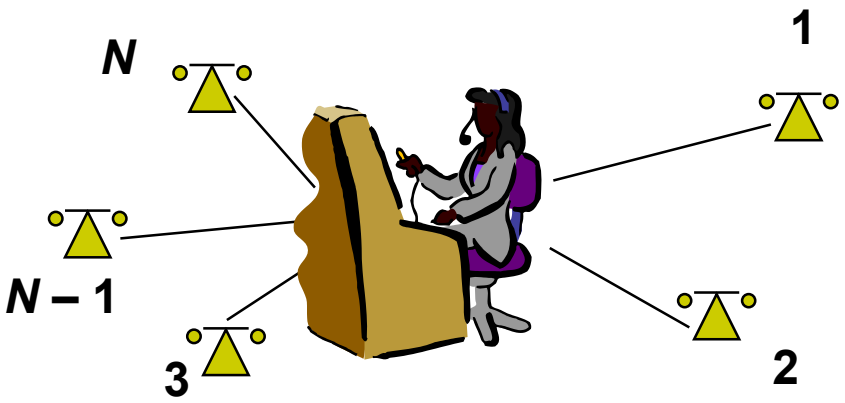
$N(N-1)/2 = 499500$

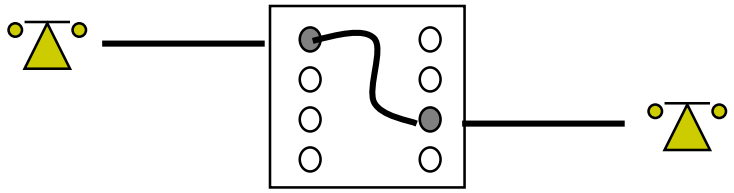
Telephone Pole Congestion



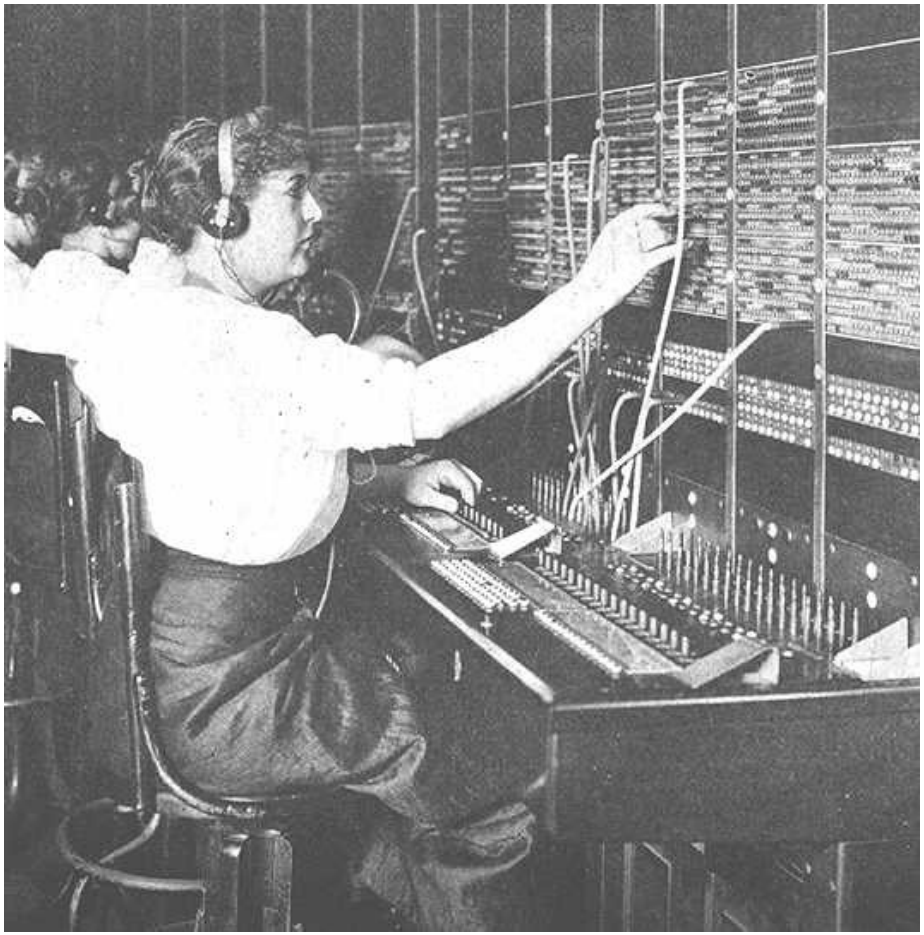
Circuit Switching

- Patchcord panel switch invented in 1877
- Operators connect users on demand
  - Establish *circuit* to allow electrical current to flow from inlet to outlet
- Only *N* connections required to central office



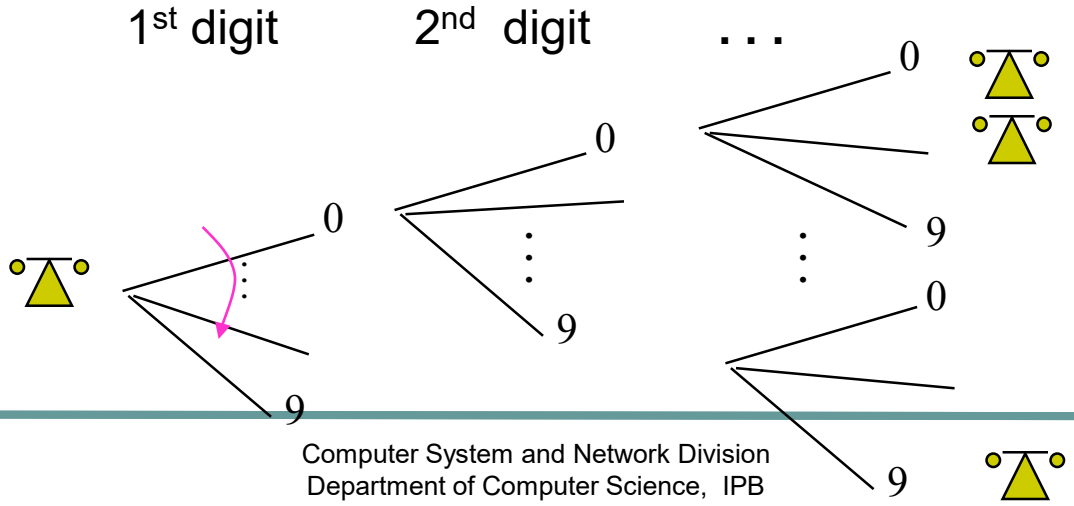


Manual Switching



Strowger Switch

- Human operators intelligent & flexible
  - But expensive and not always discreet
- Strowger invented automated switch in 1888
  - Each current pulse advances wiper by 1 position
  - User dialing controls connection setup
- Decimal telephone numbering system
- Hierarchical network structure simplifies routing
  - Area code, exchange (CO), station number



Strowger Switch

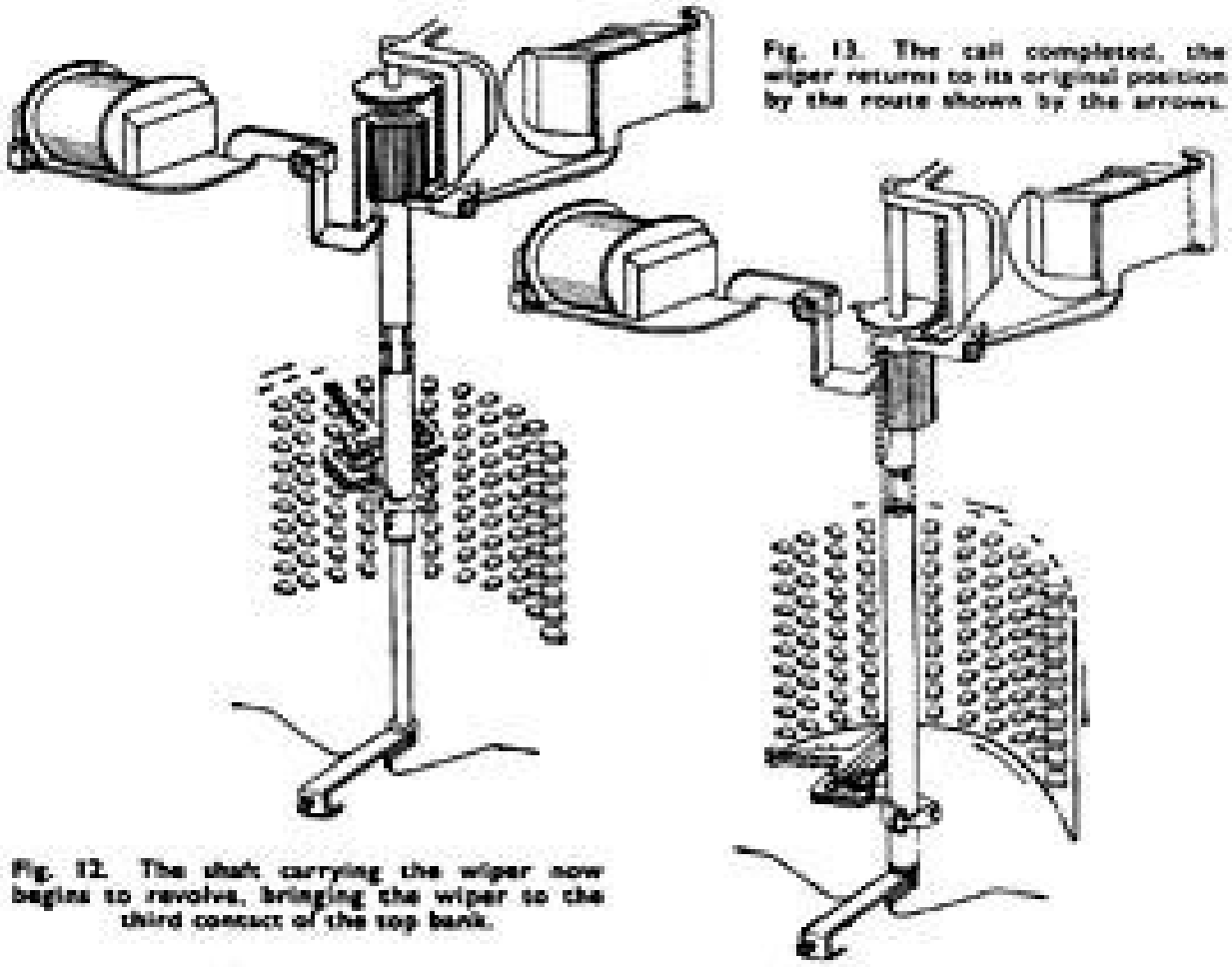
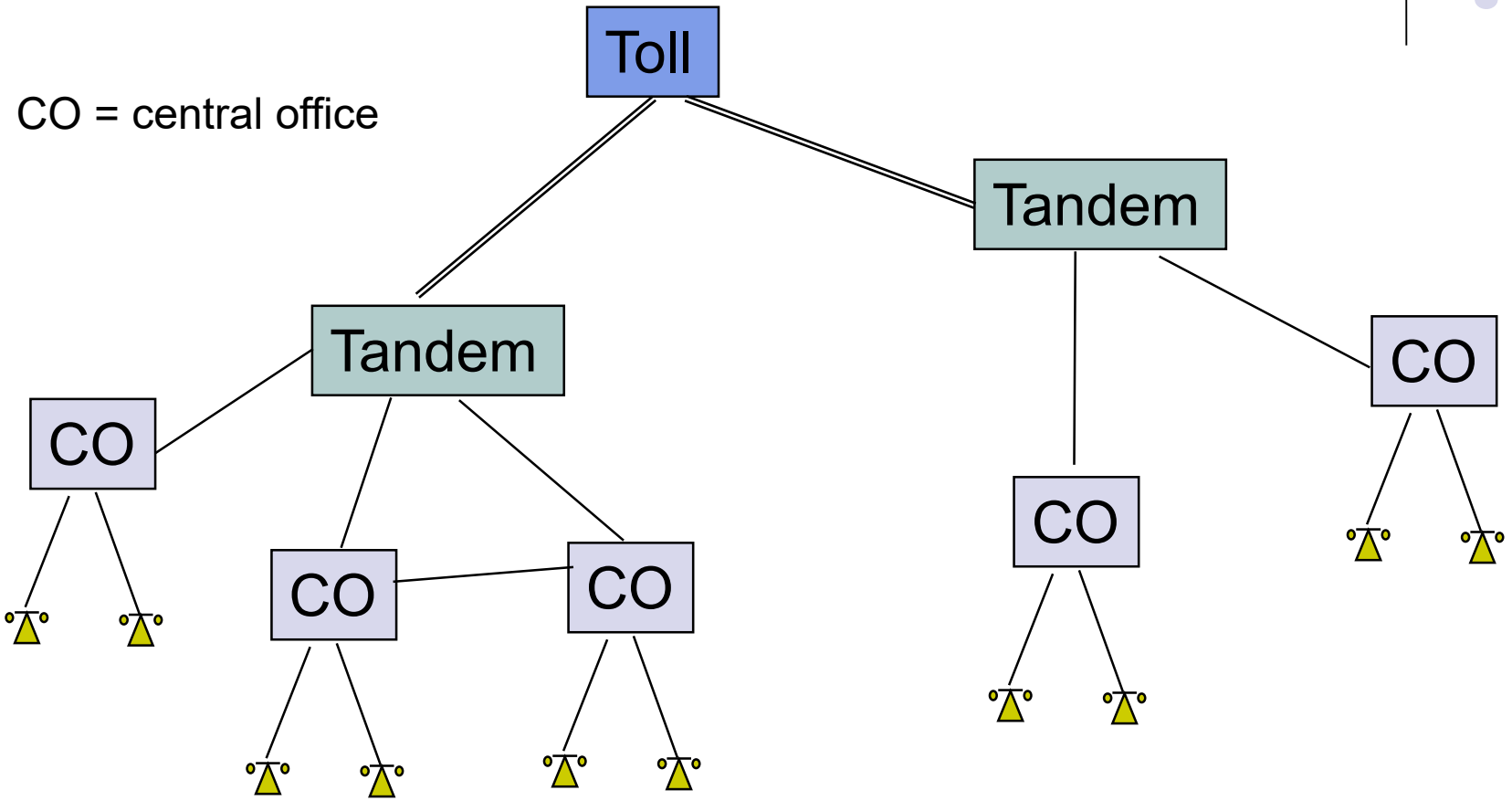


Fig. 12. The shaft carrying the wiper now begins to revolve, bringing the wiper to the third contact of the top bank.

Fig. 13. The call completed, the wiper returns to its original position by the route shown by the arrows.

Hierarchical Network Structure



Telephone subscribers connected to local CO (central office)

Tandem & Toll switches connect CO's

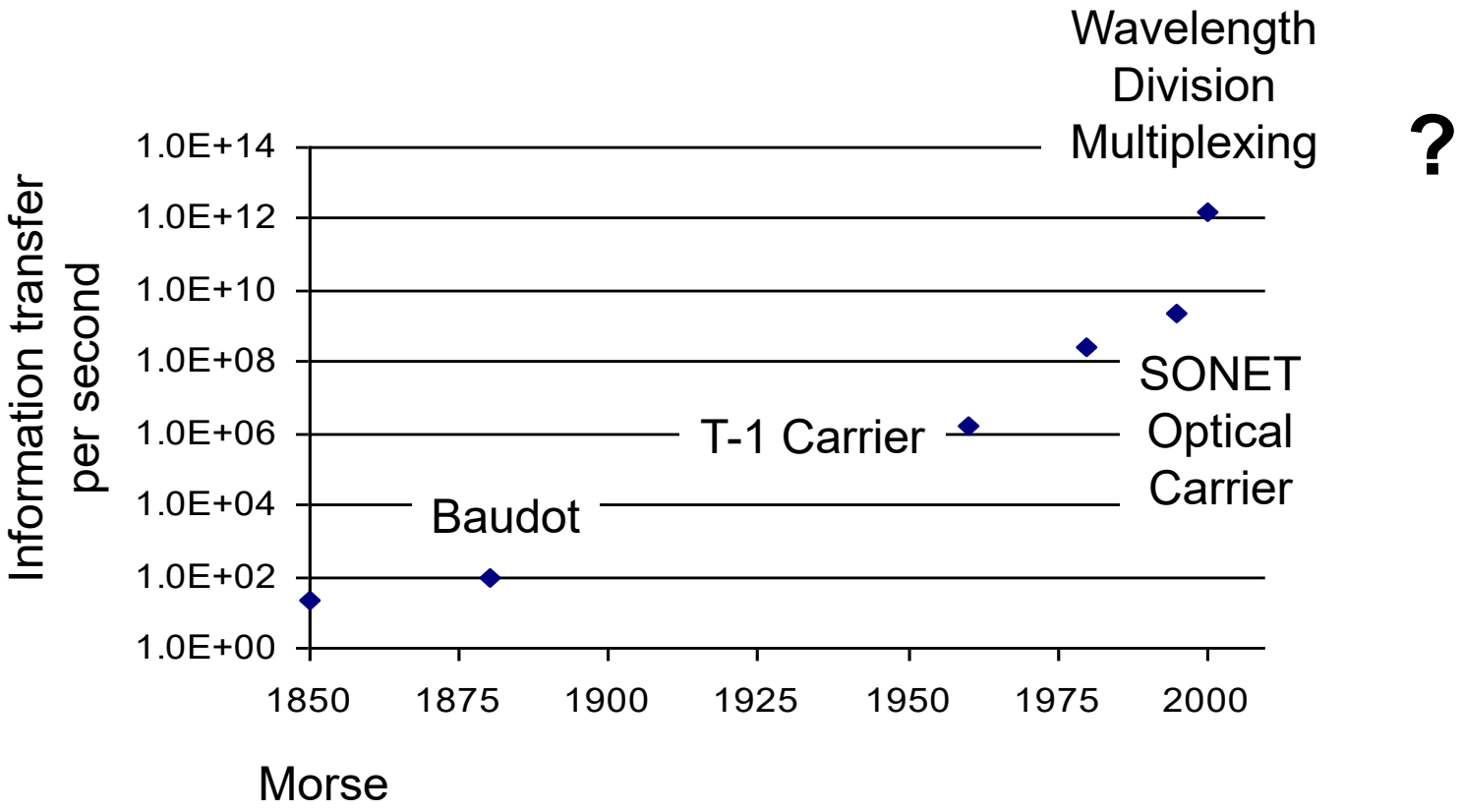
Digitization of Telephone Network



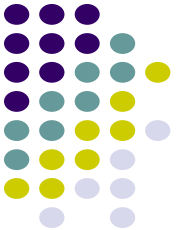
- Pulse Code Modulation digital voice signal
  - Voice gives 8 bits/sample x 8000 samples/sec = 64×10^3 bps
- Time Division Multiplexing for digital voice
  - T-1 multiplexing (1961): 24 voice signals = 1.544×10^6 bps
- Digital Switching (1980s)
  - Switch TDM signals without conversion to analog form
- Digital Cellular Telephony (1990s)
- Optical Digital Transmission (1990s)
  - One OC-192 optical signal = 10×10^9 bps
  - One optical fiber carries 160 OC-192 signals = 1.6×10^{12} bps!

All digital transmission, switching, and control

Digital Transmission Evolution



Elements of Telephone Network Architecture



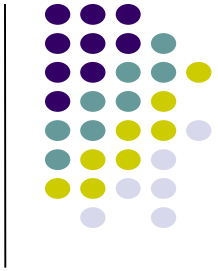
- Digital transmission & switching
  - Digital voice; Time Division Multiplexing
- Circuit switching
  - User signals for call setup and tear-down
  - Route selected during connection setup
  - End-to-end connection across network
  - Signaling coordinates connection setup
- Hierarchical Network
  - Decimal numbering system
  - Hierarchical structure; simplified routing; scalability
- Signaling Network
  - Intelligence inside the network

Computer Network Evolution Overview



- *1950s*: Telegraph technology adapted to computers
- *1960s*: Dumb terminals access shared host computer
  - SABRE airline reservation system
- *1970s*: Computers connect directly to each other
  - ARPANET packet switching network
  - TCP/IP internet protocols
  - Ethernet local area network
- *1980s & 1990s*: New applications and Internet growth
  - Commercialization of Internet
  - E-mail, file transfer, web, P2P, . . .
  - Internet traffic surpasses voice traffic

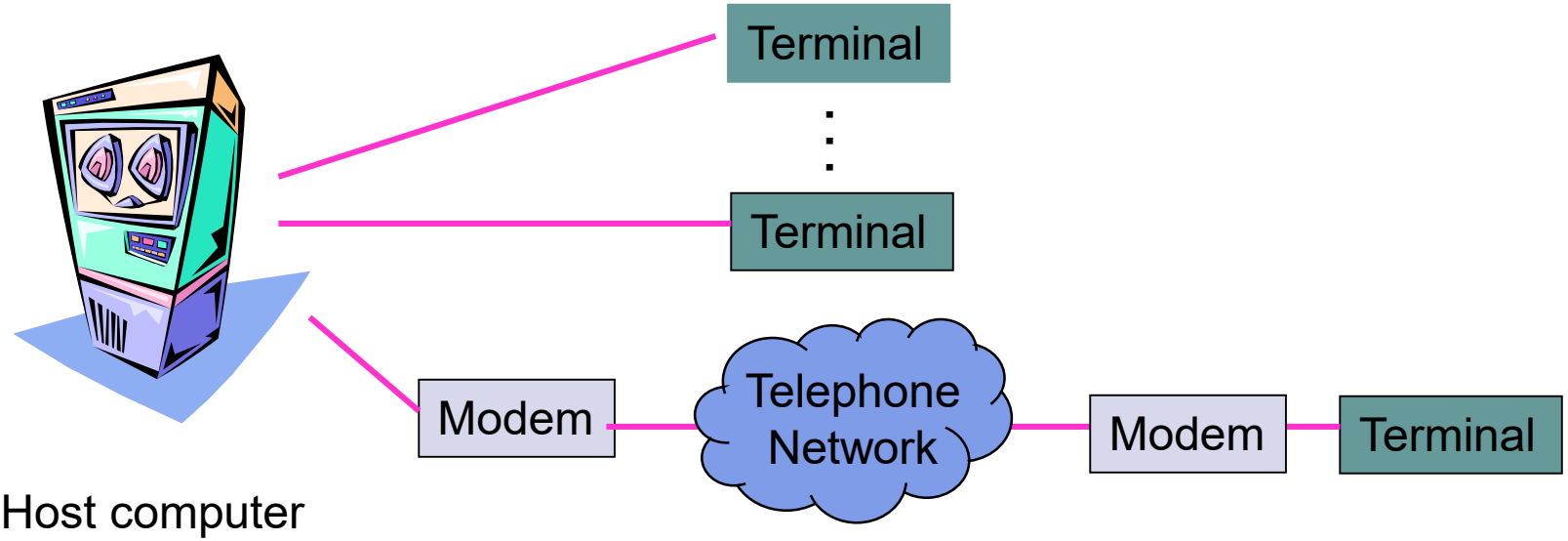
What is a protocol?



- Communications between computers requires very specific unambiguous rules
- A protocol is a set of rules that governs how two or more communicating parties are to interact
  - Internet Protocol (IP)
  - Transmission Control Protocol (TCP)
  - HyperText Transfer Protocol (HTTP)
  - Simple Mail Transfer Protocol (SMTP)

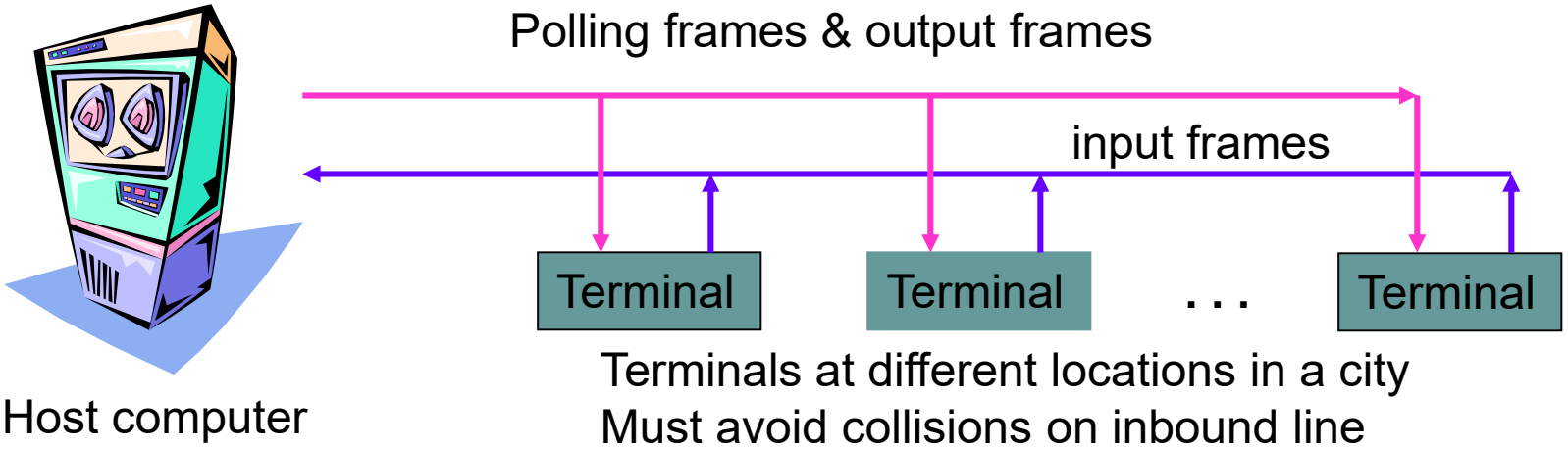
Terminal-Oriented Networks

- Early computer systems very expensive
- Time-sharing methods allowed multiple terminals to share local computer
- Remote access via telephone modems



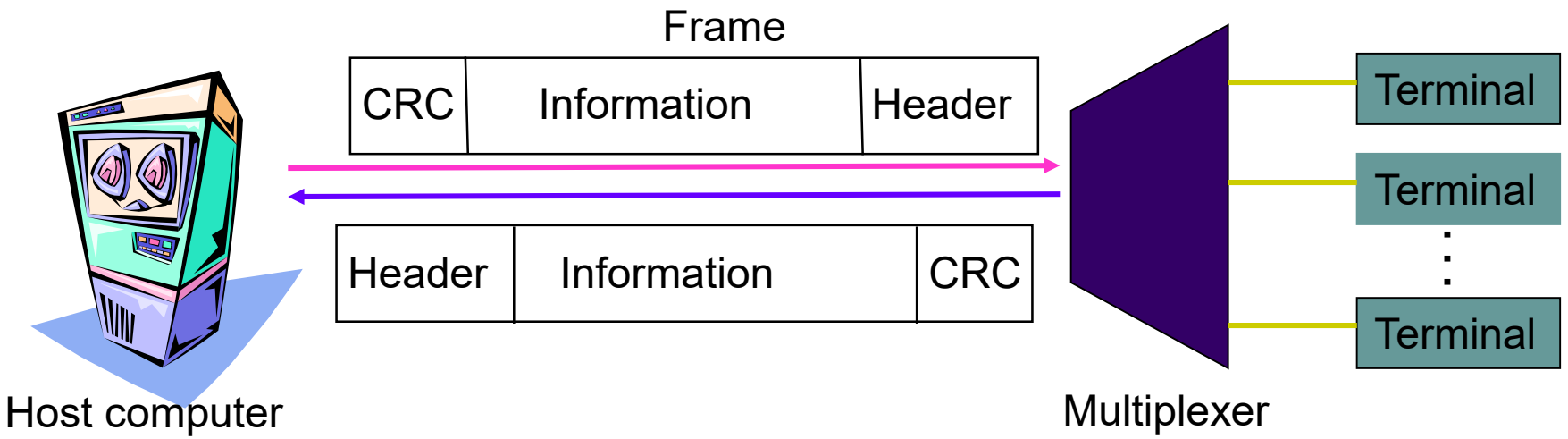
Medium Access Control

- Dedicated communication lines were expensive
- Terminals generated messages sporadically
- Frames carried messages to/from attached terminals
- Address in frame header identified terminal
- *Medium Access Controls* for sharing a line were developed
- Example: Polling protocol on a multidrop line



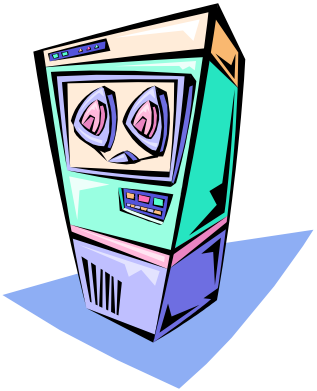
Statistical Multiplexing

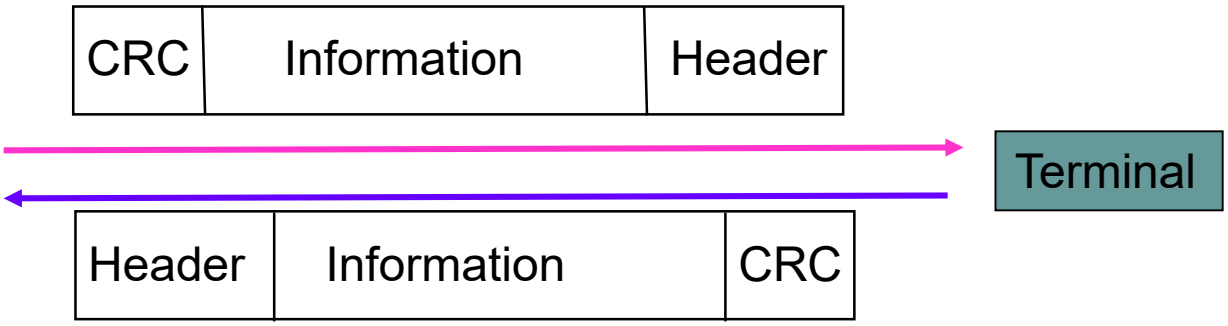
- Statistical multiplexer allows a line to carry *frames* that contain messages to/from multiple terminals
- Frames are buffered at *multiplexer* until line becomes available, i.e. store-and-forward
- *Address* in frame header identifies terminal
- Header carries other *control* information



Error Control Protocol

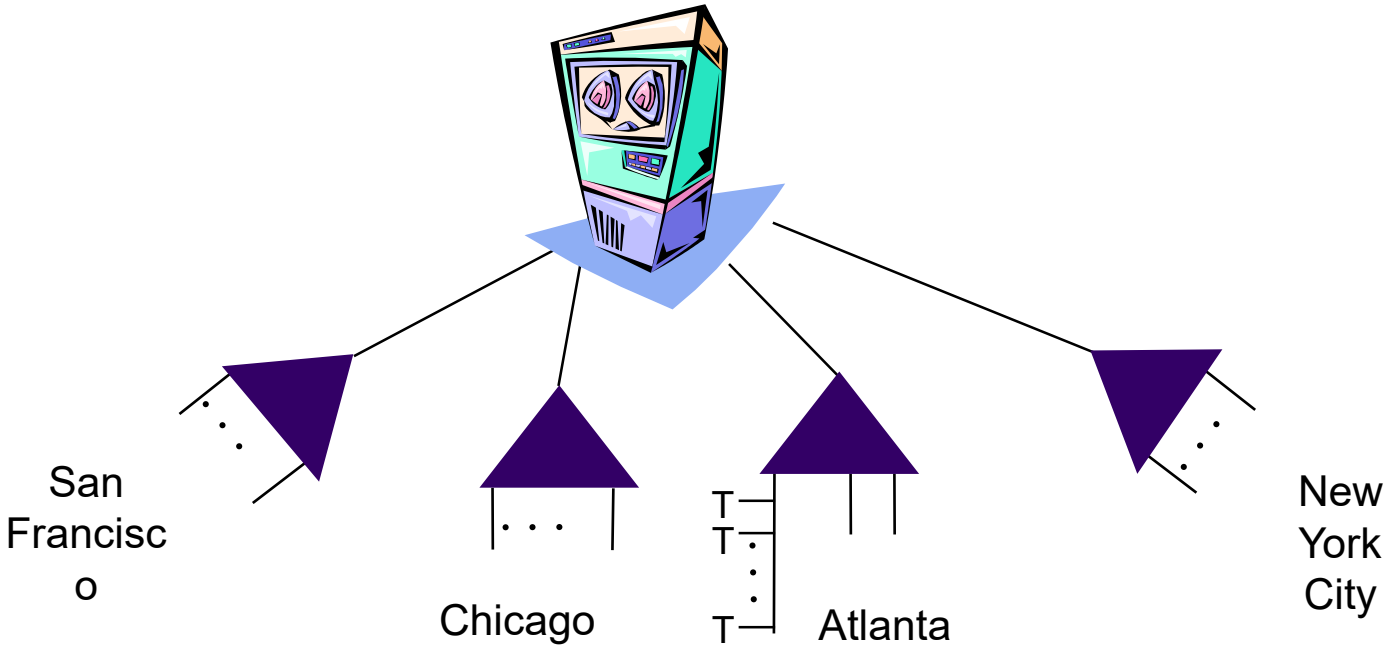
- Communication lines introduced errors
- Error checking codes used on frames
  - “Cyclic Redundancy Check” (CRC) calculated based on frame header and information payload, and appended
  - Header also carries ACK/NAK control information
- Retransmission requested when errors detected





Tree Topology Networks

- National & international terminal-oriented networks
- Routing was very simple (to/from host)
- Each network typically handled a single application

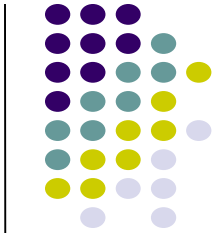


Computer-to-Computer Networks



- As cost of computing dropped, terminal-oriented networks viewed as too inflexible and costly
- Need to develop flexible computer networks
  - Interconnect computers as required
  - Support many applications
- Application Examples
  - File transfer between arbitrary computers
  - Execution of a program on another computer
  - Multiprocess operation over multiple computers

Packet Switching



- Network should support multiple applications
  - Transfer arbitrary message size
  - Low delay for interactive applications
  - But in store-and-forward operation, long messages induce high delay on interactive messages
- Packet switching introduced
  - Network transfers packets using store-and-forward
  - Packets have maximum length
  - Break long messages into multiple packets
- ARPANET testbed led to many innovations

ARPANET Packet Switching

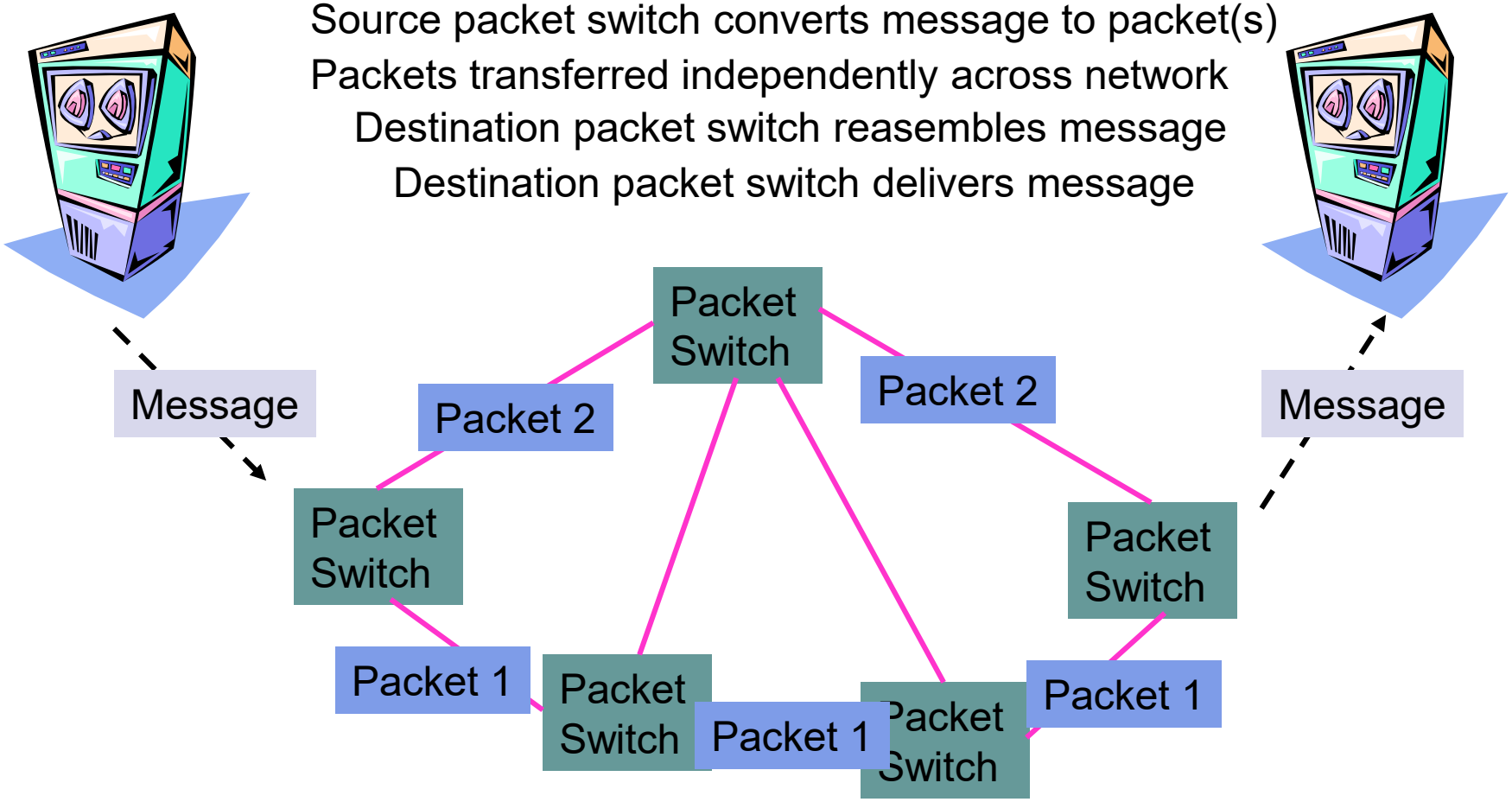
Host generates message

Source packet switch converts message to packet(s)

Packets transferred independently across network

Destination packet switch reasembles message

Destination packet switch delivers message



ARPANET Routing

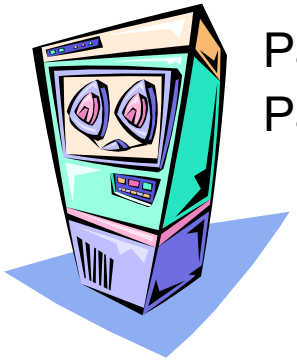
Routing is highly nontrivial in mesh networks

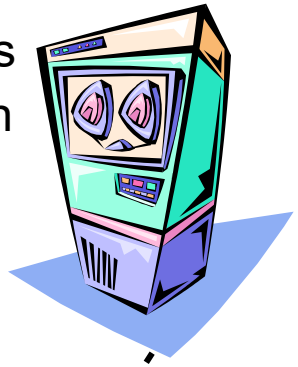
No connection setup prior to packet transmission

Packets header includes source & destination addresses

Packet switches have table with next hop per destination

Routing tables calculated by packet switches using distributed algorithm





Packet Switch

Packet Switch

Packet Switch

Packet Switch

Packet Switch

Hdr Packet

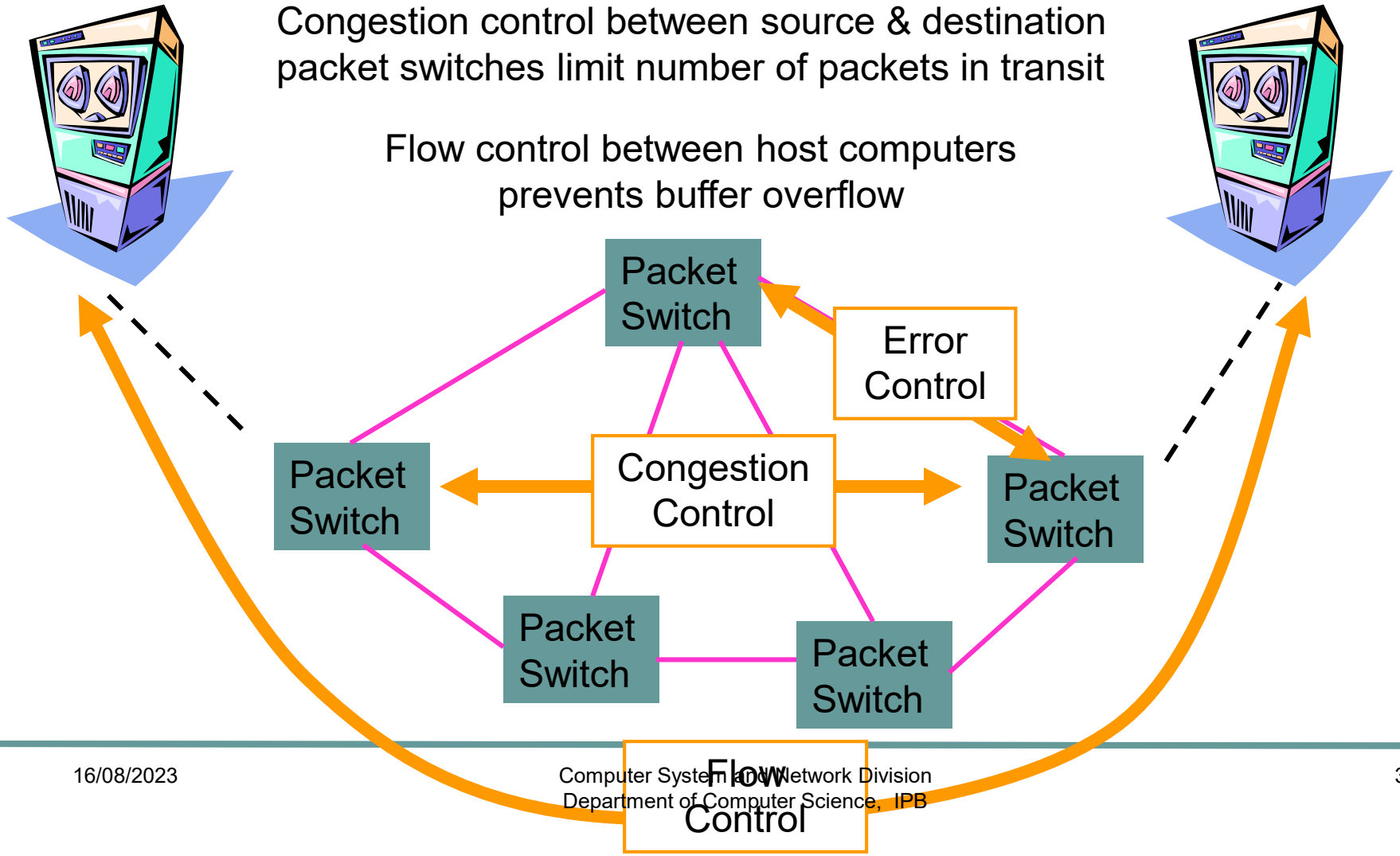
Dest:	Next Hop:
xyz	abc
wvr	edf

Other ARPANET Protocols

Error control between adjacent packet switches

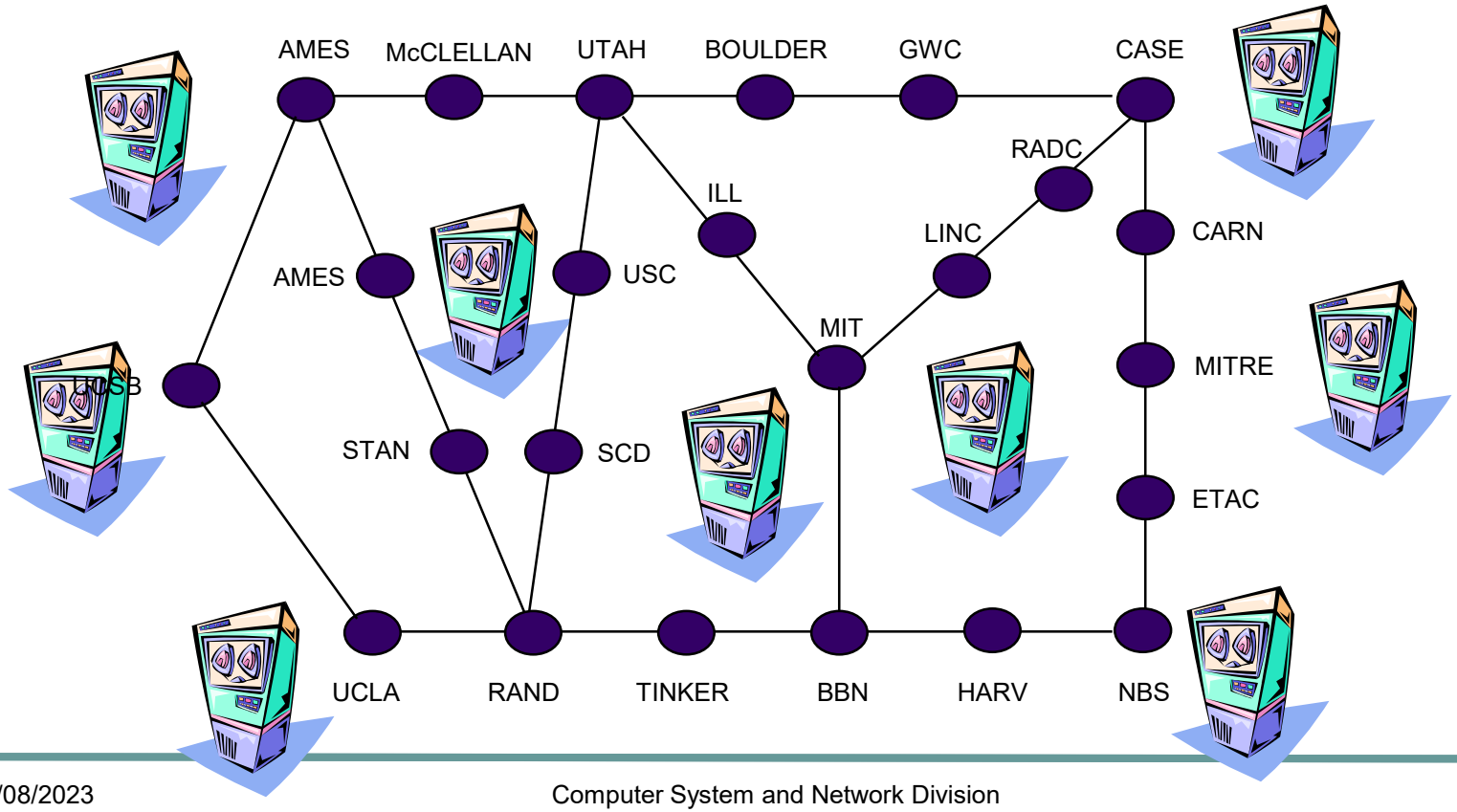
Congestion control between source & destination packet switches limit number of packets in transit

Flow control between host computers prevents buffer overflow



ARPANET Applications

- ARPANET introduced many new applications
- Email, remote login, file transfer, …



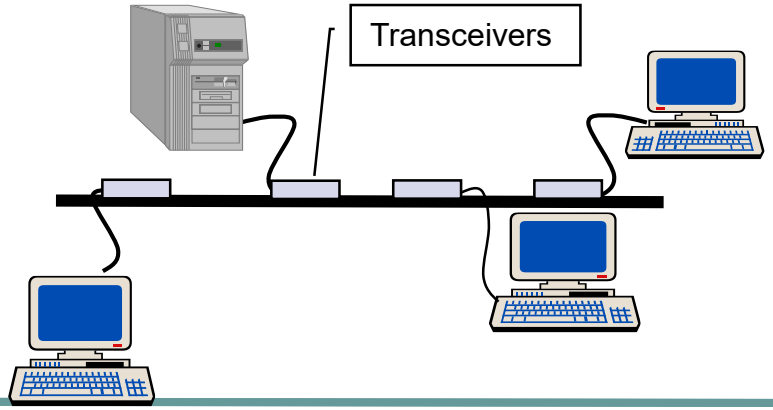
Ethernet Local Area Network



- In 1980s, affordable workstations available
- Need for low-cost, high-speed networks
  - To interconnect local workstations
  - To access local shared resources (printers, storage, servers)
- Low cost, high-speed communications with low error rate possible using coaxial cable
- Ethernet is the standard for high-speed wired access to computer networks

Ethernet Medium Access Control

- Network interface card (NIC) connects workstation to LAN
- Each NIC has globally unique address
- Frames are broadcast into coaxial cable
- NICs listen to medium for frames with their address
- Transmitting NICs listen for collisions with other stations, and abort and reschedule retransmissions



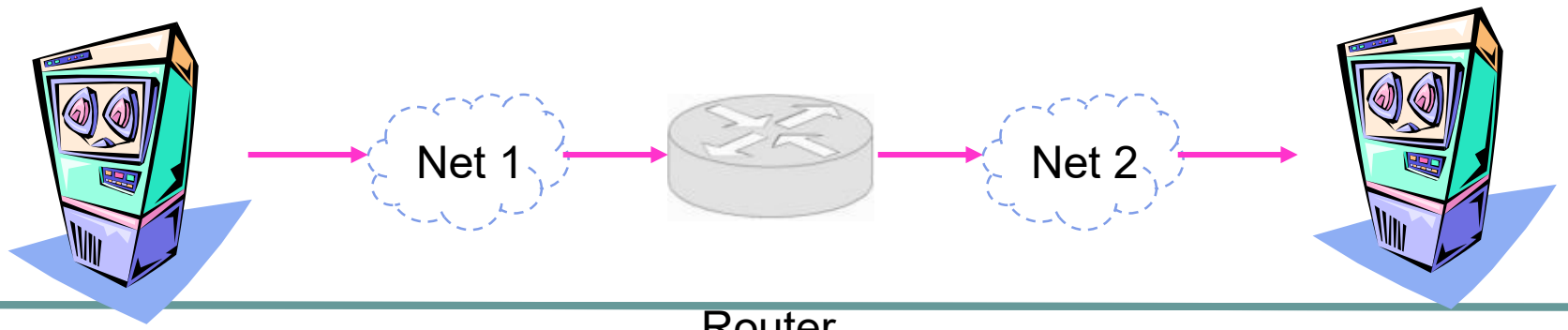
The Internet



- Different network types emerged for data transfer between computers
- ARPA also explored packet switching using satellite and packet radio networks
- Each network has its protocols and is possibly built on different technologies
- *Internetworking protocols* required to enable communications between computers attached to *different* networks
- ***Internet***: a network of networks

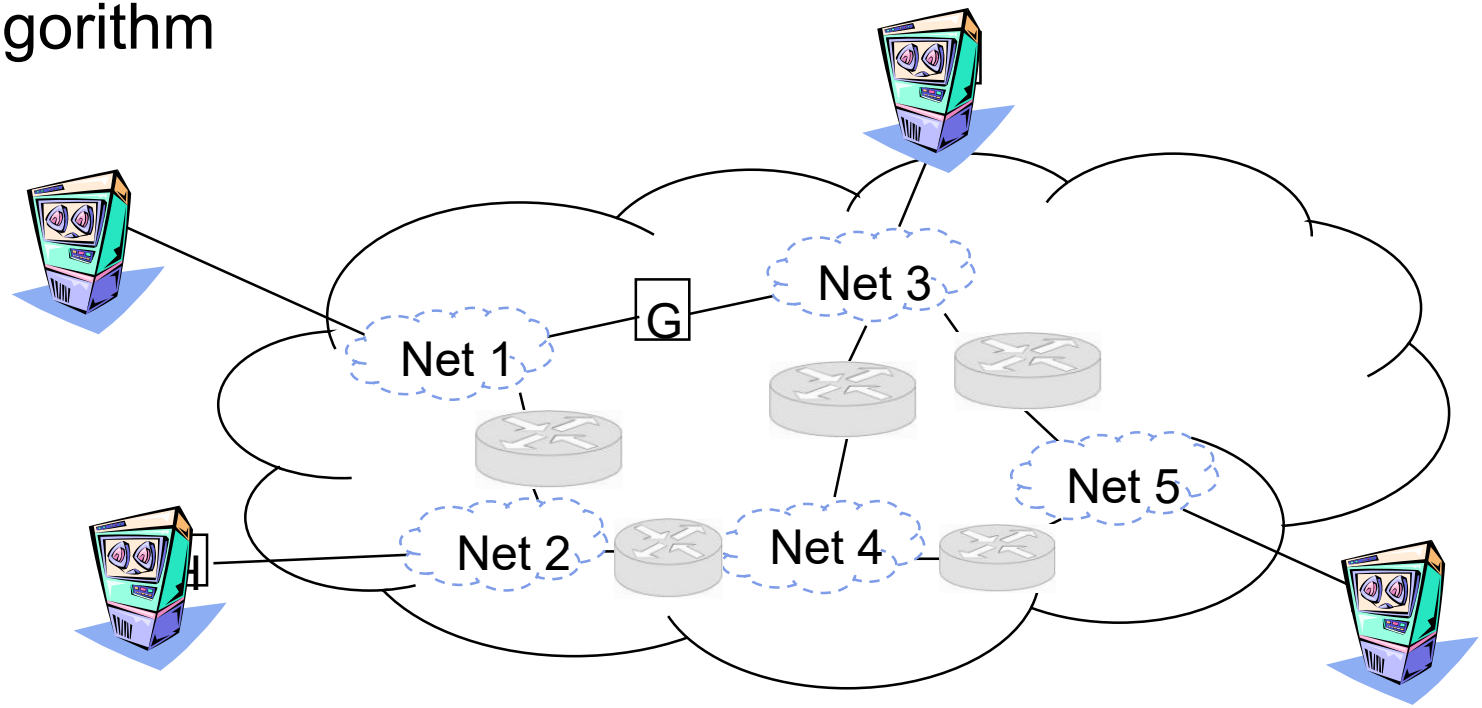
Internet Protocol (IP)

- *Routers (gateways)* interconnect different networks
- Host computers prepare IP packets and transmit them over their attached network
- Routers forward IP packets across networks
- *Best-effort* IP transfer service, no retransmission



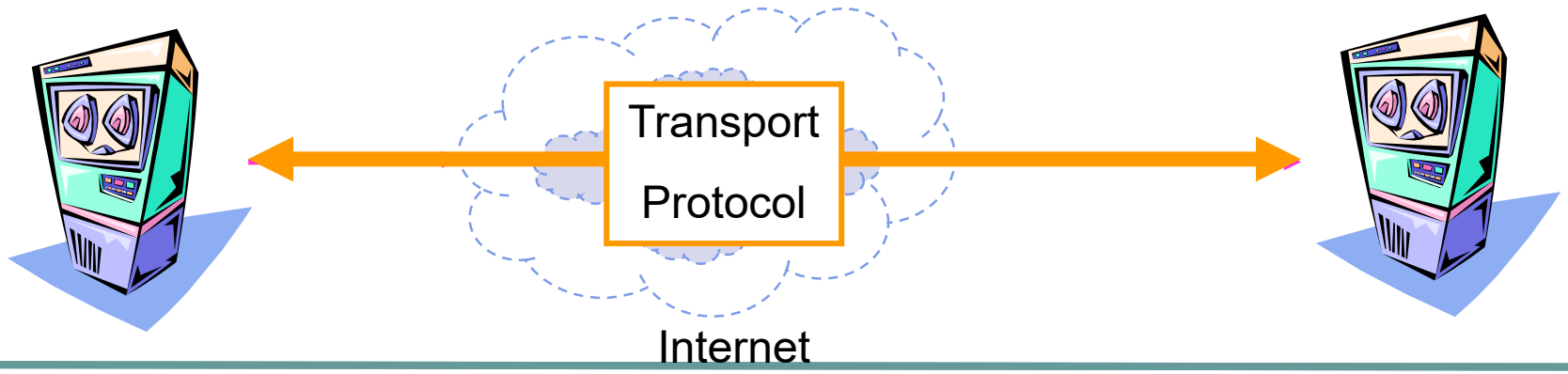
Addressing & Routing

- Hierarchical address: Net ID + Host ID
- IP packets routed according to Net ID
- Routers compute routing tables using distributed algorithm

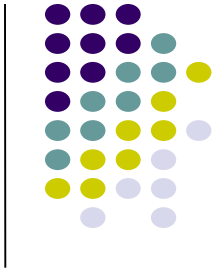


Transport Protocols

- Host computers run two transport protocols on top of IP to enable process-to-process communications
- *User Datagram Protocol* (UDP) enables best-effort transfer of individual block of information
- *Transmission Control Protocol* (TCP) enables reliable transfer of a stream of bytes

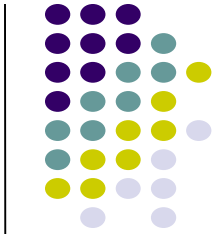


Names and IP Addresses



- Routing is done based on 32-bit IP addresses
- Dotted-decimal notation
  - 128.100.11.1
- Hosts are also identified by name
  - Easier to remember
  - Hierarchical name structure
  - tesla.comm.utoronto.edu
- Domain Name System (DNS) provided conversion between names and addresses

Internet Applications



- All Internet applications run on TCP or UDP
- TCP: HTTP (web); SMTP (e-mail); FTP (file transfer; telnet (remote terminal)
- UDP: DNS, RTP (voice & multimedia)
- TCP & UDP incorporated into computer operating systems
- Any application designed to operate over TCP or UDP will run over the Internet!!!

Elements of Computer Network Architecture



- *Digital transmission*
- Exchange of *frames* between adjacent equipment
  - Framing and error control
- *Medium access control* regulates sharing of broadcast medium.
- *Addresses* identify attachment to network or internet.
- Transfer of *packets* across a packet network
- Distributed calculation of *routing tables*

Elements of Computer Network Architecture



- *Congestion control* inside the network
- *Internetworking* across multiple networks using routers
- *Segmentation and reassembly* of messages into packets into and out of a network or internetwork
- *End-to-end transport protocols* for process-to-process communications
- *Applications* that build on the transfer of messages between computers.

Trends in Network Evolution



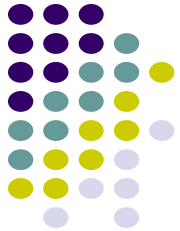
- It's all about services
  - Building networks involves huge expenditures
  - Services that generate revenues drive the network architecture
- Current trends
  - Packet switching vs. circuit switching
  - Multimedia applications
  - End of trust
  - Networking *is* a business

Packet vs. Circuit Switching



- Architectures appear and disappear over time
  - Telegraph (message switching)
  - Telephone (circuit switching)
  - Internet (packet switching)
- Trend towards packet switching at the edge
  - IP enables rapid introduction of new applications
  - New cellular voice networks packet-based
  - IP supports *real-time* voice and telephone network will gradually be replaced
  - However, large packet flows easier to manage by circuit-like methods

Optical Circuit Switching



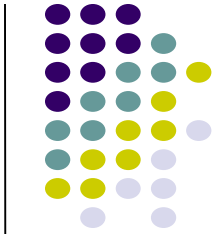
- Optical signal transmission over fiber can carry huge volumes of information (Tbps)
- Optical signal processing very limited
  - Optical logic circuits bulky and costly
  - Optical packet switching will not happen soon
- Optical-to-Electronic conversion is expensive
  - Maximum electronic speeds << Tbps
  - Parallel electronic processing & high expense
- Thus trend towards optical circuit switching in the core

Multimedia Applications



- Trend towards digitization of *all* media
- Digital voice standard in cell phones
- Music cassettes replaced by CDs and MP3's
- Digital cameras replacing photography
- Video: digital storage and transmission
  - Analog VCR cassettes largely replaced by DVDs
  - Analog broadcast TV to be replaced by digital TV
  - VCR cameras/recorders to be replaced by digital video recorders and cameras
- High-quality network-based multimedia applications now feasible

End of Trust



- Security Attacks
  - Spam
  - Denial of Service attacks
  - Viruses
  - Impersonators
- Firewalls & Filtering
  - Control flow of traffic/data from Internet
- Protocols for privacy, integrity and authentication

P2P and Overlay Networks



- Client resources under-utilized in client-server
- Peer-to-Peer applications enable sharing
  - Napster, Gnutella, Kazaa
  - Processing & storage (SETI@home)
  - Information & files (MP3s)
  - Creation of virtual distributed servers
- P2P creates transient overlay networks
  - Users (computers) currently online connect directly to each other to allow sharing of their resources
  - Huge traffic volumes a challenge to network management
  - Huge opportunity for new businesses

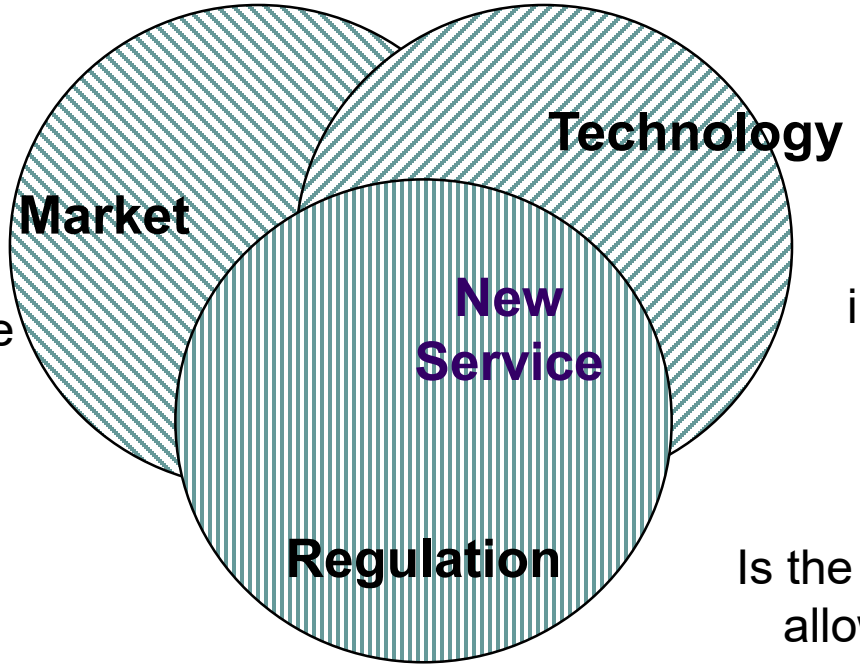
Operations, Administration, Maintenance, and Billing



- Communication like transportation networks
  - Traffic flows need to be monitored and controlled
  - Tolls have to be collected
  - Roads have to be maintained
  - Need to forecast traffic and plan network growth
- Highly-developed in telephone network
  - Entire organizations address OAM & Billing
  - Becoming automated for flexibility & reduced cost
- Under development for IP networks

Success Factors for New Services

- Technology not only factor in success of a new service
- Three factors considered in new telecom services



Can there be demand for the service?

Can it be implemented cost-effectively?

Is the service allowed?

Transmission Technology



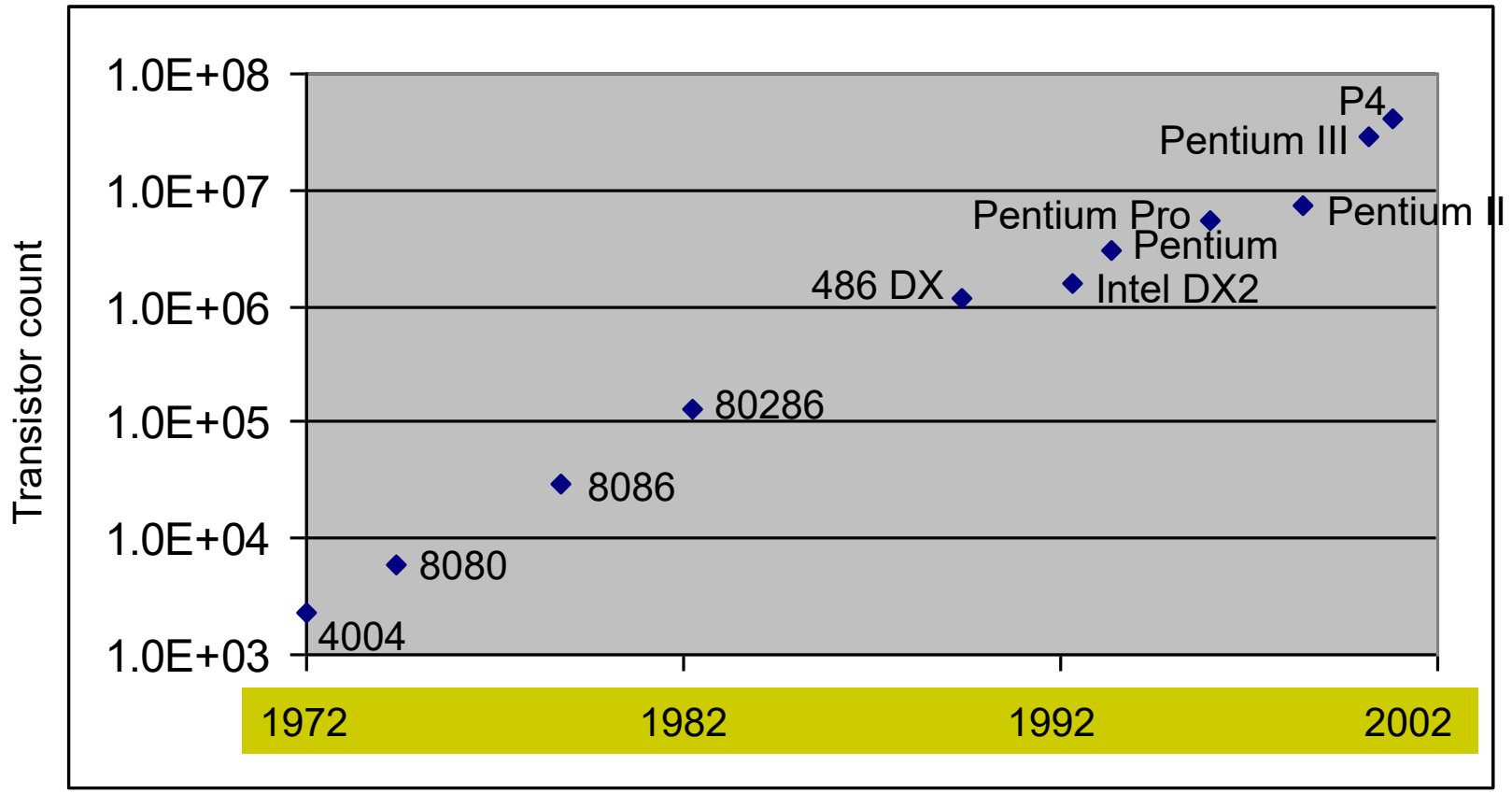
- Relentless improvement in transmission
- High-speed transmission in copper pairs
  - DSL Internet Access
- Higher call capacity in cellular networks
  - Lower cost cellular phone service
- Enormous capacity and reach in optical fiber
  - Plummeting cost for long distance telephone
- Faster and more information intensive applications

Processing Technology

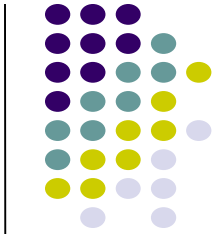


- Relentless improvement in processing & storage
- **Moore's Law**: doubling of transistors per integrated circuit every two years
- RAM: larger tables, larger systems
- Digital signal processing: transmission, multiplexing, framing, error control, encryption
- Network processors: hardware for routing, switching, forwarding, and traffic management
- Microprocessors: higher layer protocols and applications
- Higher speeds and higher throughputs in network protocols and applications

Moore's Law

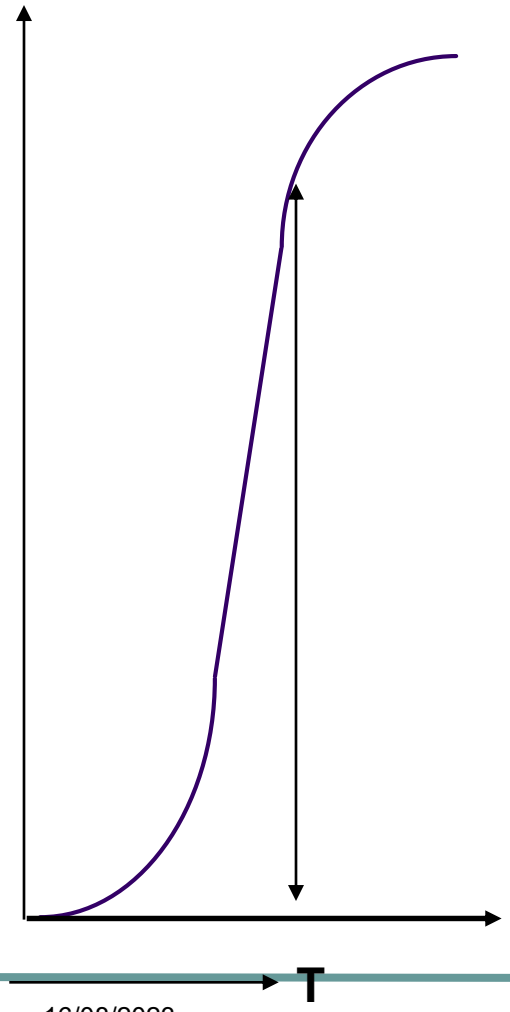

Transistor count
1.0E+08
1.0E+07
1.0E+06
1.0E+05
1.0E+04
1.0E+03
1972
1982
1992
2002
P4
Pentium III
Pentium II
Pentium Pro
Pentium
Intel DX2
486 DX
80286
8086
8080
4004

Market



- *The network effect*: usefulness of a service increases with size of community
  - **Metcalfe's Law**: usefulness is proportional to the square of the number of users
  - Phone, fax, email, ICQ, …
- *Economies of scale*: per-user cost drops with increased volume
  - Cell phones, PDAs, PCs
  - Efficiencies from multiplexing
- *S-curve*: growth of new service has S-shaped curve, challenge is to reach the critical mass

The S Curve



Service Penetration & Network Effect

- Telephone: T=30 years
  - city-wide & inter-city links
- Automobile: T=30 years
  - roads
- Others
  - Fax
  - Cellular & cordless phones
  - Internet & WWW
  - Napster and P2P

Regulation & Competition



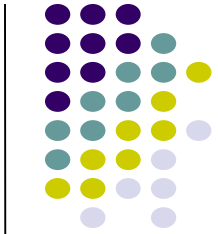
- Telegraph & Telephone originally monopolies
  - Extremely high cost of infrastructure
  - Profitable, predictable, slow to innovate
- Competition feasible with technology advances
  - Long distance cost plummeted with optical tech
  - Alternative local access through cable, wireless
  - Radio spectrum: auctioned vs. unlicensed
- Basic connectivity vs. application provider
  - Tussle for the revenue-generating parts

Standards



- New technologies very costly and risky
- Standards allow players to share risk and benefits of a new market
  - Reduced cost of entry
  - Interoperability and network effect
  - Compete on innovation
  - Completing the value chain
    - Chips, systems, equipment vendors, service providers
- Example
  - 802.11 wireless LAN products

Standards Bodies



- Internet Engineering Task Force
  - Internet standards development
  - Request for Comments (RFCs): www.ietf.org
- International Telecommunications Union
  - International telecom standards
- IEEE 802 Committee
  - Local area and metropolitan area network standards
- Industry Organizations
  - MPLS Forum, WiFi Alliance, World Wide Web Consortium